

STUDENT PERFORMANCE ANALYSIS

Data Acquisition, Cleaning and Exploratory Data Analysis

Week 1 Data Science Assignment

Project Detail	Information
Dataset	Students Performance in Exams
Dataset File	StudentsPerformance.csv
Records	1,000 students
Variables	8 columns
Tools Used	Python, Pandas, Matplotlib, Seaborn, Google Colab
Analysis Type	Data Cleaning and Exploratory Data Analysis (EDA)

Prepared as an academic data analysis report

1. Introduction

Student performance analysis is useful for understanding how learners perform across different subjects and identifying relationships among academic scores. This project uses the Students Performance in Exams dataset to demonstrate a complete beginner-level data preparation and exploratory analysis workflow using Python.

The main objectives are to acquire the dataset, inspect its structure, check data quality, identify missing and duplicate records, summarize numerical variables, and visualize useful patterns in student scores.

2. Dataset Description

The dataset contains 1,000 student records and 8 variables. It includes demographic and educational attributes together with mathematics, reading and writing scores.

Column	Type	Description
gender	Categorical	Student gender
race/ethnicity	Categorical	Student group
parental level of education	Categorical	Parent/guardian education level
lunch	Categorical	Lunch program category
test preparation course	Categorical	Whether preparation was completed
math score	Numerical	Mathematics score
reading score	Numerical	Reading score
writing score	Numerical	Writing score

3. Data Acquisition

The dataset was obtained from Kaggle as a publicly available CSV dataset. It was uploaded to Google Colab and loaded into a Pandas DataFrame for analysis.

```
import pandas as pd

df = pd.read_csv("/content/StudentsPerformance.csv")
print(df.head())
print(df.shape)
print(df.info())
```

The initial inspection confirmed 1,000 rows and 8 columns. Five columns are categorical and three columns are numerical score variables.

4. Data Cleaning and Quality Checks

Data cleaning was performed without deleting original student records. The dataset was checked for missing values, duplicate rows and data types. Since no missing or duplicate records were found, no row removal or imputation was required.

```
print(df.isnull().sum())
print("Number of duplicate rows:", df.duplicated().sum())
print(df.dtypes)
```

Quality Check	Result	Action Taken
Missing values	0 in all 8 columns	No imputation required
Duplicate rows	0	No duplicate removal required
Data types	5 object + 3 integer columns	Suitable for analysis
Original records	1,000	Kept unchanged

5. Descriptive Statistics

The following summary was obtained for the three numerical score variables.

Statistic	Math	Reading	Writing
Count	1000	1000	1000
Mean	66.09	69.17	68.05
Standard deviation	15.16	14.60	15.20
Minimum	0	17	10
25th percentile	57.00	59.00	57.75
Median	66.00	70.00	69.00
75th percentile	77.00	79.00	79.00
Maximum	100	100	100

6. Exploratory Data Analysis

Exploratory Data Analysis was carried out to understand score levels, compare average performance across groups, and identify relationships between subjects.

6.1 Average Score by Subject

Reading has the highest overall average score at 69.17, followed by writing at 68.05 and mathematics at 66.09.

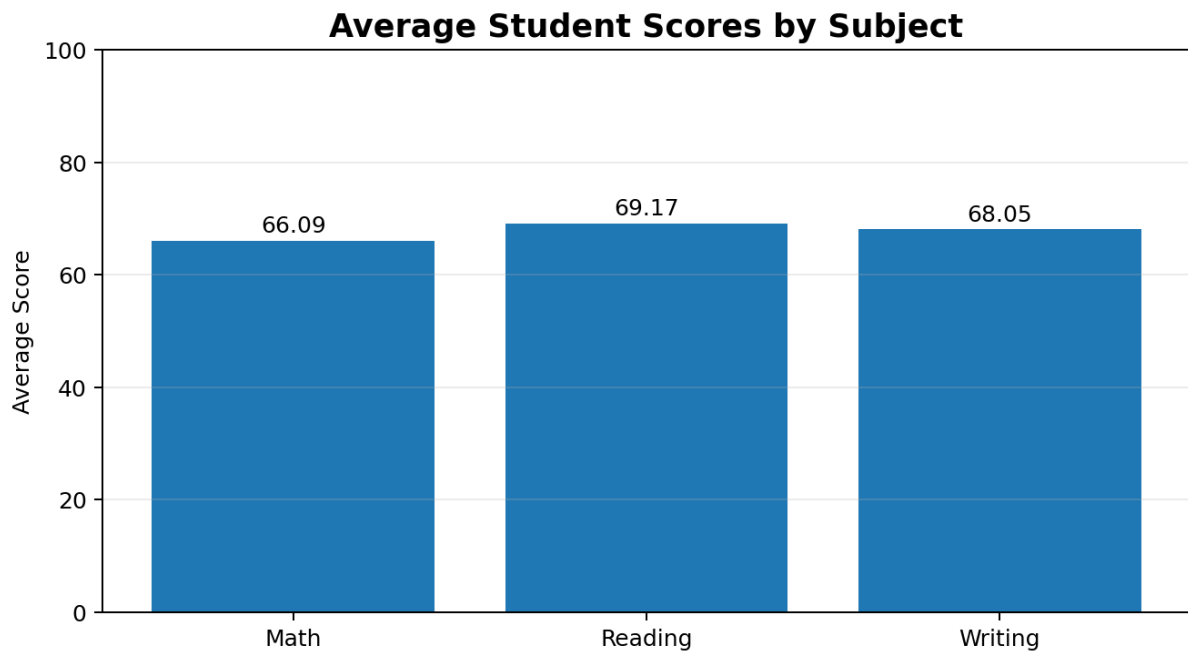


Figure 1. Average student score by subject.

6.2 Average Score by Gender

The gender-wise comparison shows that female students have higher average reading and writing scores, while male students have a higher average mathematics score in this dataset.

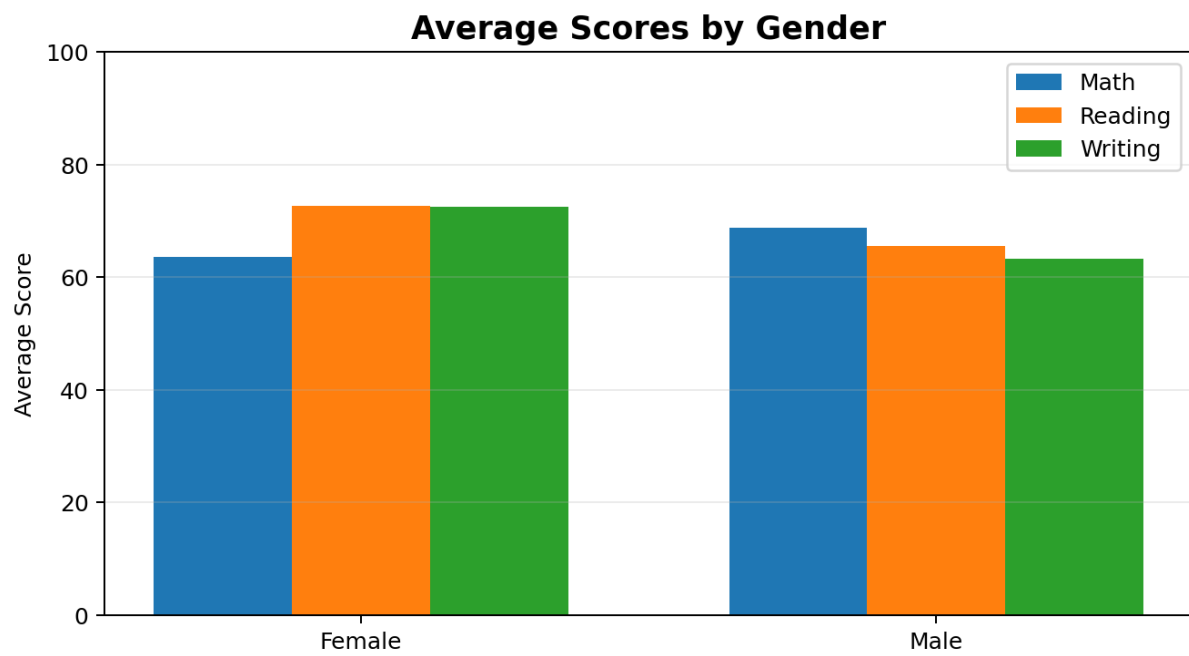


Figure 2. Average scores by gender.

6.3 Correlation Between Subjects

Correlation analysis shows strong positive relationships among all three subjects. Reading and writing have the strongest relationship (0.95), while mathematics is also strongly related to reading (0.82) and writing (0.80).

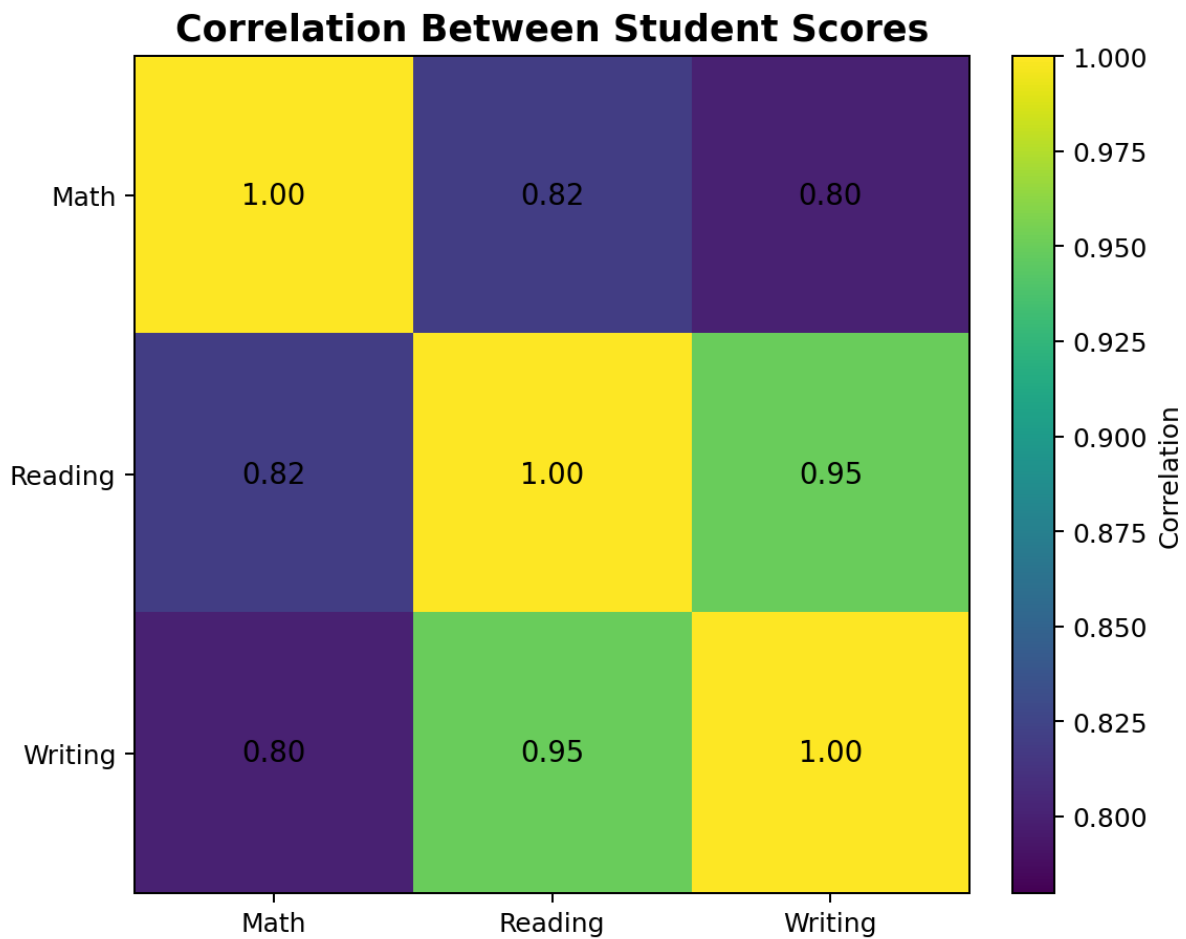


Figure 3. Correlation heatmap for mathematics, reading and writing scores.

7. Key EDA Insights

- The dataset contains 1,000 students with 8 variables and is complete with no missing values.
- No duplicate records were detected, so the original dataset did not require duplicate removal.
- Reading has the highest overall mean score (69.17), while mathematics has the lowest (66.09).
- Reading and writing scores have a very strong positive correlation (0.95).
- Mathematics also has strong positive correlations with reading (0.82) and writing (0.80).
- The results indicate that students who perform well in one academic subject generally tend to perform well in the other subjects.
- The analysis describes associations in the available dataset; it does not establish causal relationships.

8. Rationale for Cleaning Methods

Missing-value checks were necessary because incomplete records can affect averages, correlations and visualizations. Duplicate checks were performed to ensure that repeated records did not unintentionally influence the analysis. Data types were inspected so that categorical variables remained categorical and numerical scores could be summarized and

correlated correctly. Because the dataset had no missing or duplicate records, applying unnecessary transformations would not improve data quality and could change the original evidence.

9. Limitations

The dataset is a cross-sectional student performance dataset and does not contain a time variable. Therefore, it cannot be used to establish how student performance changes over time. The correlations reported here describe association, not cause and effect.

10. Conclusion

This project successfully completed the Week 1 data preparation and exploratory analysis workflow. The Students Performance in Exams dataset was acquired, loaded into Python, checked for missing values and duplicates, summarized statistically, and explored through multiple visualizations. The analysis found strong positive relationships among mathematics, reading and writing scores, with reading showing the highest overall average. The cleaned dataset is suitable for further analysis or future predictive modeling.

Appendix: Main Analysis Code

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("/content/StudentsPerformance.csv")

# Structure and quality checks
print(df.head())
print(df.shape)
print(df.info())
print(df.isnull().sum())
print("Number of duplicate rows:", df.duplicated().sum())

# Numerical summary
print(df[["math score", "reading score", "writing score"]].describe())

# Correlation heatmap
correlation = df[["math score", "reading score", "writing score"]].corr()
sns.heatmap(correlation, annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Correlation Between Student Scores")
plt.show()
```